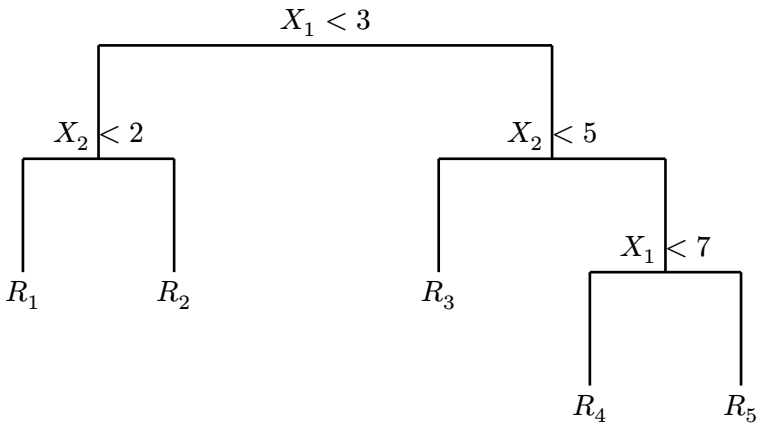
The University of Texas at Austin
In-Class Work 8

M339G Predictive Analytics

November 03, 2026

Regression Trees.

Problem 8.1. You are given the following regression tree with predictors X_1 and X_2, where the inequality at each split describes the left branch:



Which of the following correctly describes region R_4?

a. $\{X \mid X_1 < 7, X_2 \geq 5\}$
b. $\{X \mid X_1 \geq 7, X_2 \geq 5\}$
c. $\{X \mid 3 \leq X_1 < 7, X_2 < 5\}$
d. $\{X \mid 3 \leq X_1 < 7\}$
e. $\{X \mid 3 \leq X_1 < 7, X_2 \geq 5\}$

Solution. The correct answer is **e.**
To reach R_4, an observation must go right at the root ($X_1 \geq 3$), right at the second split ($X_2 \geq 5$), and left at the third split ($X_1 < 7$). Combining these conditions gives

$$R_4 = \{X \mid 3 \leq X_1 < 7, X_2 \geq 5\}.$$

Choice (A) ignores the root split, and choice (D) ignores the split on X_2.

Problem 8.2. A regression tree is fit using recursive binary splitting. Which of the following statements about the resulting regions $R_1, \ldots, R_J$ is true?

a. Two regions may overlap, so an observation can belong to more than one region.
b. The number of regions equals the number of internal (splitting) nodes.
c. Each predictor can be used to split at most once in the tree.
d. The regions are non-overlapping boxes whose union is the entire predictor space.
e. A boundary between two regions may be a diagonal line such as $X_1 + X_2 = 4$.

Solution. The correct answer is **d.**
Each split divides an existing box into two boxes along a single predictor, so the regions are always non-overlapping (high-dimensional) rectangles that together cover the whole predictor space.

- a. is false: every observation follows exactly one path to one terminal node.
- b. is false: a binary tree with J terminal nodes has $J - 1$ internal nodes.
- c. is false: the same predictor can be split on repeatedly, as X_1 is in Problem 1.
- e. is false: every split has the form $X_j < s$, so all boundaries are parallel to the axes.

Problem 8.3. A regression tree has been fit to a training data set. Consider the following statements:

I. The predicted response for a new observation is the mean of the training responses in the terminal node it falls into.
II. Two new observations that fall into the same terminal node receive the same predicted response, even if their predictor values differ.
III. The predicted response for a new observation can exceed the largest response in the training data.

Determine which of the statements are true.

a. I only
b. II only
c. I and II only
d. I and III only
e. I, II, and III

Solution. The correct answer is **c.**
- I is true: the tree predicts with the mean training response in each region.
- II is true: the prediction depends only on the region, so it is constant within each region.
- III is false: each prediction is a mean of training responses, so it lies between the smallest and the largest training response.

Problem 8.4. You are given the following training data with one predictor X and response Y:

X	1	2	3	4	5
Y	2	3	9	10	11

The first split of a regression tree is chosen by recursive binary splitting from the candidate cutpoints $s \in \{1.5, 2.5, 3.5, 4.5\}$, with the left branch $X < s$. Determine the cutpoint chosen.

(A) 1.5
(B) 2.5
(C) 3.5
(D) 4.5
(E) It cannot be determined without a validation set.

Solution. The correct answer is **b.**
Recursive binary splitting chooses the cutpoint that minimizes the total RSS of the two resulting nodes, where each node predicts with its own mean.

- $s = 1.5$: $\{2\}$ and $\{3, 9, 10, 11\}$ with mean 8.25. RSS $= 0 + 38.75 = 38.75$.
- $s = 2.5$: $\{2, 3\}$ with mean 2.5 and $\{9, 10, 11\}$ with mean 10. RSS $= 0.5 + 2 = 2.5$.
- $s = 3.5$: $\{2, 3, 9\}$ with mean $14/3$ and $\{10, 11\}$ with mean 10.5. RSS $= 28.67 + 0.5 = 29.17$.

- $s = 4.5$: $\{2, 3, 9, 10\}$ with mean 6 and $\{11\}$. RSS $= 50 + 0 = 50$.

The smallest RSS occurs at $s = 2.5$, which separates the small responses from the large ones. No validation set is needed, since the split is chosen on training RSS.

Problem 8.5. Consider the following statements about recursive binary splitting for regression trees:

I. It is a greedy algorithm: at each step it makes the best split for that step, without looking ahead to splits that might lead to a better tree later.
II. Among all trees with the same number of terminal nodes, it is guaranteed to produce the one with the smallest training RSS.
III. At each step, it considers every predictor and every possible cutpoint for that predictor.

Determine which of the statements are true.

a. I and III only
b. I only
c. II only
d. I and II only
e. I, II, and III

Solution. The correct answer is **a.**

- I is true: this is what makes the algorithm greedy.
- II is false: because the algorithm is greedy, a split that looks poor now but enables much better splits later can be missed, so the final tree need not minimize training RSS among trees of its size.
- III is true: at each step the algorithm searches over all predictors X_j and cutpoints s for the split that minimizes the RSS.